



VIVEKANANDA INSTITUTE OF PROFESSIONAL STUDIES - TECHNICAL CAMPUS

Grade A++ Accredited Institution by NAAC

NBA Accredited for MCA Programme; Recognized under Section 2(f) by UGC;
Affiliated to GGSIP University, Delhi; Recognized by Bar Council of India and AICTE
An ISO 9001:2015 Certified Institution

ASSIGNMENT 2

Subject: Unsupervised Learning

Subject Code : ML-465T

Semester : 7th

Branch : CSE

S. No.	Question	Marks	CO	PO	BL
1	Differentiate between Undercomplete, Regularized, and Denoising Autoencoders. Provide real-world applications of each.	4	3	1,2,4	2,3
2	What is Mode Collapse in GANs? Explain its causes and methods used to overcome it.	4	2,3	2,3,5	4
3	Explain novelty detection and outlier detection. How do they differ? Provide an example of each.	4	2,3	3,5,12	3,5
4	Describe Kernel Density Estimation (KDE). Explain how KDE helps in density estimation with a suitable diagram.	4	1,4	6,7,8	1,2
5	Explain the architecture and training algorithm of Kohonen Self-Organizing Feature Maps (SOFM).	4	4	6,8,12	4,5